Update on “How much do high schools matter?: Effects on college choice beyond achievement”

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Project Overview

- Goal: estimate the causal effect of schools on higher-education outcomes.
- Central question: do schools affect major choice through achievement gains, or also through non-score channels?
- A key practical input is the computation of assignment probabilities in the Chilean centralized school-assignment system.
- Empirical strategy proposes a combination of reduced-form school-effect estimation with a structural model of demand for majors.

Conceptual Framework

- Assume a constant-effects model for school value added:

$$Y_{ij} = \nu_j + \epsilon_i$$

- This implies school effects can be written relative to the average school effect:

$$Y_i = \beta_0 + \sum_{j=1}^{J} \beta_j D_{ij} + \epsilon_i$$

- The core challenge is selection: students attending a school may differ systematically in unobservables.

Identification from Centralized Assignment

- In centralized assignment, students submit ranked preferences and schools rank students by priority.
- Randomness enters through lottery tie-breakers.
- Define student type as the combination of preferences and priorities:

$$\theta_i \equiv (\succ_i, \rho_i)$$

- Under "equal treatment of equals" (ETE), students with the same type face the same assignment probabilities. (Abdulkadiroğlu et al. 2017)

$$\epsilon_i \perp\!\!\!\perp Z_i \mid \theta_i$$

- Other (fixed under randomization) variables do not affect the assignment. Key example: ϵ_i.

Assignment Risk as a Control Strategy

- Saturating on exact student type is often infeasible because preference-priority cells are too sparse.
- A practical alternative is to use assignment probabilities p_i as propensity scores.

  (Abdulkadiroğlu et al. 2017; Rosenbaum and Rubin 1983)
- As a result we can write instead:

$$\epsilon_i \perp\!\!\!\perp Z_i \mid p_i$$

- This motivates "risk-controlled" designs that condition on school-assignment risk.

Example: Common Risk Without Common Type

Setup

- Applicants: $\{1, 2, 3, 4, 5\}$
- Schools: $\{a, b, c\}$, one seat each
- Priority:
  - 5 highest at c
  - 2 highest at b

Preferences

$$\begin{aligned} 1: &\quad a \succ b \\ 2: &\quad a \succ b \\ 3: &\quad a \\ 4: &\quad c \succ a \\ 5: &\quad c \end{aligned}$$

DA implication

- 5 gets c
- 1, 2, 3, 4 compete for a
- none has an advantage at a

$$P(Z_{ia} = 1) = 0.25 \qquad \forall i \in \{1, 2, 3, 4\}$$

Takeaway

- No two applicants share the same full type
- Full-type conditioning gives one stratum per applicant
- But applicants 1–4 share the same propensity score for a

From Offers to Attendance

- Offers Z_i are exogenous conditional on assignment risk p_i.
- To move from offers to actual attendance D_i, the project invokes a conditional-independence argument:

$$\epsilon_i \perp\!\!\!\perp D_i \mid p_i$$

- Intuition: sorting occurs at the application stage, while post-offer compliance is driven by idiosyncratic shocks. (Dale and Krueger 2002)
- This delivers a tractable framework for school-specific reduced-form estimation. (Angrist et al. 2023)

Reduced-Form School Effects

- Example outcomes:
  - standardized math achievement,
  - STEM enrollment in higher education.
- Yet also interested in other fields; outcomes such as financial aid application.
- Proposed specifications:

$$Y_i^{\text{MATH}} = b_0 + \sum_j b_j D_{ij} + X_i' B + b_{\text{pre}} Y_{\text{pre},i} + g(p_i) + \epsilon_i$$

$$Y_i^{\text{STEM}} = c_0 + \sum_j c_j D_{ij} + X_i' C + g(p_i) + \eta_i$$

- Under Angrist et al. (2024), these correspond to an RC VAM specification and a risk-controlled OLS.

Why Major Choice May Reflect More Than Scores

- If math achievement affects STEM enrollment, then school effects on STEM can operate through:
  1. a score channel, and
  2. a non-score channel.
- In that case, the reduced-form coefficient on school attendance in the STEM regression mixes both channels.
- A potentially useful test: focus on schools with estimated zero effects on math scores.
- Non-zero STEM effects for those schools would be evidence of non-score school effects.

Ideally: An additional score-only instrument

- In situations like these, the gold-standard for mediation analysis is having access to an instrument that affects only one of the variables (e.g. scores)
- Currently considering using exam forms (randomly assigned, might give a boost of achievement to some students), but concerned about potentially weak first-stage.

Another alternative: Structural Model of Major Choice

- Complementary approach: estimate a demand model over broad fields of study.
- Utility from field f:

$$u_{if} = \delta_{g(i)f} - \kappa_{g(i)} d_{if} + f_{g(i)f}(Y^{\text{MATH}}, Y^{\text{VERBAL}}) + \lambda_{fs} + \eta_{if}$$

- Key object of interest: λ_{fs}, a school-field preference shifter.
- This allows schools to shift preferences even after flexibly accounting for test scores.
- Similar models estimated by Abdulkadiroğlu et al. (2020) and Campos et al. (2026), but in both cases they decompose it in other (causal) objects.

Rationale behind the Structural Approach

- College ranked lists are interpreted as preference revelations under student-proposing deferred acceptance.
- Importantly, I propose to model major choice and abstract away from particular programs.
- Covariate cells are defined by macro-region, gender, and family income.

- With 12 covariate cells, 8 fields, and 200 schools, the current parameter count is around 2000 parameters.
- For reference Abdulkadiroğlu et al. (2020) estimate 720 parameters and Campos et al. (2026) around 288.
- Potentially going down on the covariate cells or the flexibility of the score function seems like a reasonable compromise.

Replicating the Chilean Assignment System

- In the Chilean context, the mechanism includes many features beyond textbook DA.
- Important institutional features include:
  - reserved spots for specific groups,
  - rollover of unused quota seats into regular seats,
  - guaranteed admission to the origin school,
  - sibling block applications,
  - transition rules that change over early implementation years.
- These details matter a lot empirically.

Current Replication Progress

- Managed to replicate above 99% of original assignments with my 4 sample cohorts. (!!)
- Using only student types and a random seed replicated about 68% of assignments.
- This highlights the importance of correctly modeling the additional institutional features.
- Around 30% of the sample in the centralized system get the same assignment regardless of the lottery number.
- Next steps: explain the remaining mismatch, and try to implement the large market approximation from Abdulkadıroğlu et al. (2022), Abdulkadiroğlu et al. (2017).

Replicating Example 1

type_id	school_name	prob
1	a	0.24880
2	a	0.25122
3	a	0.24731
4	a	0.25267
1	b	0.25122
2	b	0.74878
5	c	1.00000
1	unmatched	0.49998
3	unmatched	0.75269
4	unmatched	0.74733

Table 1: Assignment probabilities by type and school

Next Steps

- Next steps:
  - Consult about doubts with estimation.
  - Implement Empirical Bayes to reduce noise in the estimators.
  - Estimate the demand model.

Preliminary Results

Effects: math achievement

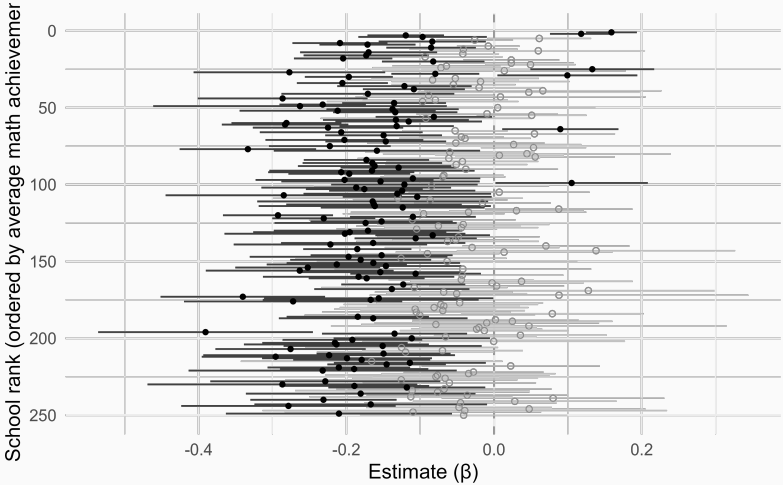

Figure 1: Estimates and confidence intervals for distribution of β_s

Effects: verbal achievement

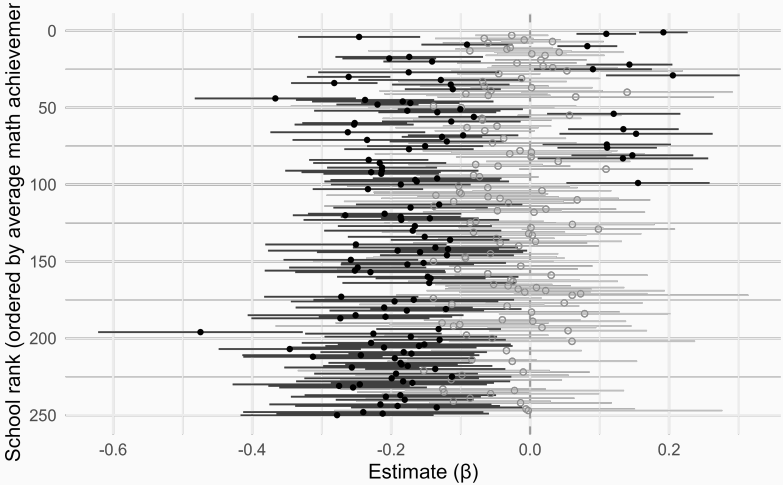

Figure 2: Estimates and confidence intervals for distribution of β_s

Effects: STEM enrollment

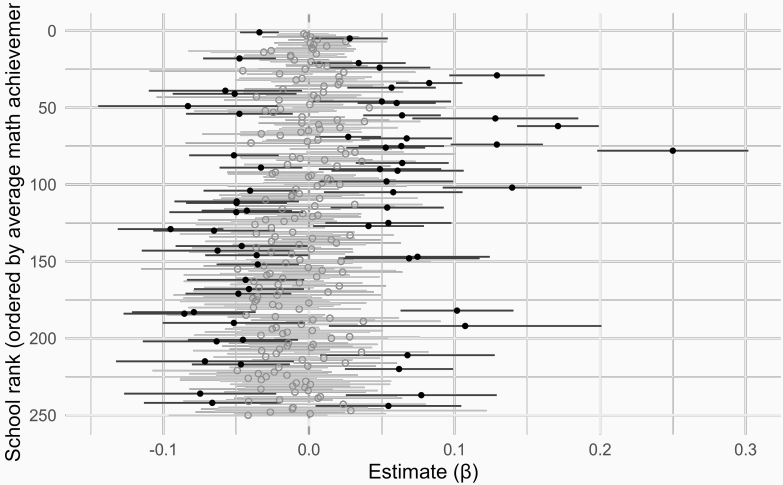

Figure 3: Estimates and confidence intervals for distribution of β_s

Effects: Correlation between math and verbal achievement

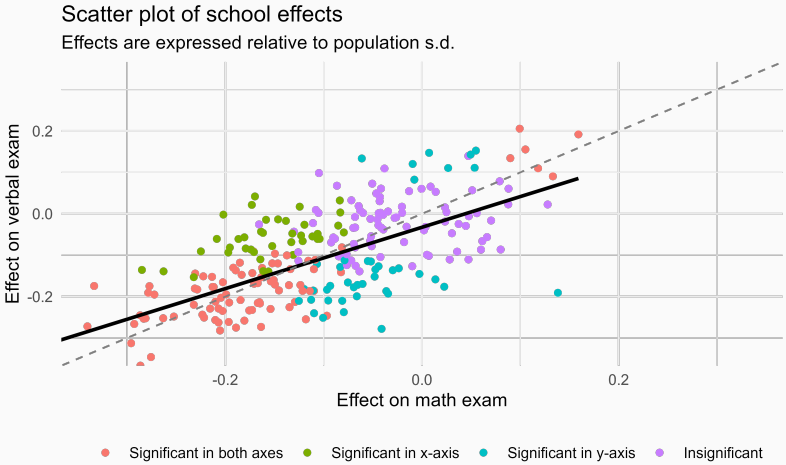

Figure 4: Estimates and confidence intervals for distribution of β_s

Effects: Correlation between math and STEM enrollment

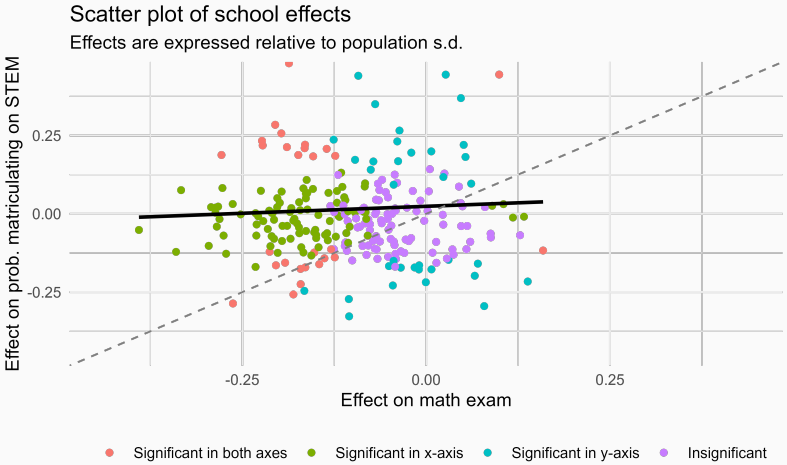

Figure 5: Estimates and confidence intervals for distribution of β_s

Thanks!

I would really appreciate your feedback !

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